

Rolling Legendre-Fenchel Rate Functions as Crypto Fragility Diagnostics

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Abstract—Scalar risk measures such as volatility, VaR, and tail-index estimates compress tail behavior into one-dimensional summaries and can obscure the empirical geometry of downside fragility. This paper studies rolling empirical Legendre-Fenchel rate-function surfaces as diagnostic state variables for cryptocurrency tail risk, not as universal event timers. The central object is a fragility coordinate: the minimum exponential-tilting cost needed to make a drawdown-consistent average return plausible in the recent window. On BTCUSD and ETHUSD daily data from 2019–2026, this coordinate is most informative for cross-regime separability. For 30-day BTC drawdowns of at least 20%, HDLDR reaches AUC 0.631 versus 0.529 for realized volatility, but the dependence-robust block bootstrap is marginal ($p = 0.064$ at a 21-day block); the significant DeLong result ($z = 4.71$, $p < 0.001$) is therefore treated as a parametric reference rather than the binding inference. Evidence is weaker for ETH and for short horizons, and incremental gains over a multivariate XGBoost benchmark are not robust under block bootstrap. The strongest mechanistic evidence comes from a one-hour Binance Futures liquidation-pressure extension using licensed 2025–2026 records: for 72-hour future liquidation pressure, adding 1h LDT features raises Spearman IC from 0.418 to 0.629 for BTC and from 0.436 to 0.622 for ETH, with positive 336-hour block-bootstrap intervals. Methodologically, a horizon-consistent threshold correction avoids a $26.9\times$ unit mismatch that otherwise makes the BTC estimator non-computable. HDLDR can be calibrated into a valid 99% VaR model, but it does not improve quantile loss relative to Gaussian VaR. Rolling rate surfaces therefore complement, rather than replace, volatility and VaR models.

Index Terms—Cryptocurrency risk, large deviation theory, rate function, tail risk, regime detection, Value-at-Risk, extreme value theory.

I. INTRODUCTION

Cryptocurrency markets are not merely high-volatility versions of traditional risky assets. They combine continuous trading, fragmented liquidity, large retail and offshore participation, and leveraged perpetual futures. These features can create feedback loops: adverse price moves trigger forced deleveraging, which can add market pressure and further deform the short-run return distribution. This motivates a view of the market as a non-stationary system whose stability landscape may change across regimes before a large drawdown is fully realized.

Standard tools summarize risk through scalar quantities such as realized volatility, Value-at-Risk (VaR), Expected Shortfall (ES), EVT tail parameters, or rolling quantiles. These quantities are important and remain natural benchmarks, but they collapse the local tail geometry into one-dimensional summaries. Two windows can have similar volatility or similar

quantiles while requiring different distributional tilts to make a large negative average return typical. The object of interest in this paper is that empirical barrier: the local exponential cost of moving from the recent return distribution to a downside drawdown state.

Large-deviation theory (LDT) provides the language for this barrier. Under ideal assumptions, a rate function controls the exponential decay of rare sample-mean probabilities [1], [2]. The present paper does not apply the theorem literally to financial returns, which are dependent, heavy-tailed, and non-stationary. Instead, it uses the rolling empirical rate function as a diagnostic state variable. When the estimated rate function falls at a drawdown-relevant return level, the recent empirical distribution requires less exponential tilting to make that drawdown typical; in economic terms, the downside barrier has become thinner.

This framing leads to an ex ante hypothesis about horizons. A rolling rate surface estimated over a multi-month window should be better suited to detecting persistent fragility across regimes than to timing abrupt one-week shocks driven by liquidity, news, or immediate volatility spikes. The empirical question is therefore not whether large-deviation features universally forecast crashes, but whether they add an interpretable cross-regime fragility signal for severe medium-horizon drawdown risk.

The main contributions are as follows:

- 1) We define rolling empirical rate surfaces as interpretable diagnostic state variables for cryptocurrency fragility, with log-sum-exp stabilization and effective-sample-size filtering.
- 2) We introduce horizon-adjusted rate-function features that align cumulative drawdown thresholds with average log-return large-deviation thresholds. The correction addresses a concrete implementability failure: using cumulative drawdown thresholds directly as daily rate-function targets renders the estimator non-computable for BTC, as documented in Table III.
- 3) We provide mechanistic evidence using one-hour Binance Futures liquidation data, testing whether rate-surface dynamics matter when the target is realized future liquidation pressure rather than a rare binary daily drawdown label.
- 4) We evaluate diagnostic cross-regime separability for medium-horizon BTC drawdowns on daily 2019–2026 data, while reporting weaker evidence for ETH and short-horizon labels.

- 5) We report mixed evidence transparently: rate surfaces can be calibrated into a valid VaR model but do not improve VaR quantile loss, and incremental daily drawdown gains over strong multivariate benchmarks are not robust under block bootstrap.

Rate surfaces are not proposed as superior crash predictors. They are interpretable fragility coordinates whose incremental value is clearest when the evaluation target is aligned with persistent stress or with realized liquidation pressure.

The rest of this paper is organized as follows. Section II reviews related work. Section III defines the estimator and features. Section IV describes the data and validation design. Section V reports empirical results. Section VI discusses interpretation and limitations. Section VII concludes.

II. RELATED WORK

Tail risk in cryptocurrency markets. VaR, ES, GARCH-family models, and EVT have been applied to crypto returns [3], [4], [5], [6]. More recent work shows that crypto tail risk is dynamic and model-sensitive: regime-switching CAViAR improves joint VaR/ES forecasts [7], dynamic POT improves Bitcoin tail forecasts [8], EWMA variants remain competitive for crypto portfolios [9], structural breaks and fat tails materially affect VaR and ES estimates [10], and EVT-copula and tail-network studies highlight cross-asset tail dependence [11], [12]. This literature provides the benchmark forecasting layer for this paper.

Large deviations and entropic risk. LDT provides a rigorous theory of rare fluctuations through cumulant generating functions and rate functions [1], [2], with exponential tilting also underlying rare-event simulation in financial engineering [13]. The entropic risk measure $\rho_\lambda(X) = \lambda \log \mathbb{E}[\exp(-X/\lambda)]$ is the CGF $\Lambda(\theta)$ evaluated at $\theta = -1/\lambda$ [14], [15]; its convex conjugate is the rate function $I(x) = \sup_\theta \{\theta x - \Lambda(\theta)\}$, which is zero at the mean and grows in the tail. The present work uses the empirical rate function as a rolling diagnostic statistic rather than invoking a new LDT theorem: a low $\hat{I}_{t,w}(x)$ at a downside level x indicates that the recent empirical distribution requires a smaller exponential tilt to place mass near that drawdown state.

Regime detection and crypto microstructure. Markov-switching and regime-dependent GARCH specifications capture changes in volatility dynamics [16], [5]. Crypto microstructure adds funding rates, perpetual-futures basis, open interest, and liquidation pressure as relevant state variables [17], [18], [19]. Liquidation clusters relate naturally to self-exciting point processes [20], [21], and crypto returns display large jumps that motivate diagnostics beyond a single volatility state [22], [23].

III. METHODOLOGY

A. Returns and Rolling Windows

Let P_t denote the daily close price and

$$r_t = \log P_t - \log P_{t-1} \quad (1)$$

the log return. For window length w , features dated t use only

$$\mathcal{W}_{t,w} = \{r_{t-w}, \dots, r_{t-1}\}. \quad (2)$$

This timing convention prevents look-ahead in all predictive tests.

B. Empirical CGF and Rate Function

For $\theta \in \Theta = [\theta_{\min}, \theta_{\max}]$, the empirical cumulant generating function (CGF) is

$$\hat{\Lambda}_{t,w}(\theta) = \log \left(\frac{1}{w} \sum_{r_i \in \mathcal{W}_{t,w}} \exp(\theta r_i) \right). \quad (3)$$

It is computed using

$$\hat{\Lambda}_{t,w}(\theta) = m + \log \left(\frac{1}{w} \sum_i \exp(\theta r_i - m) \right), \quad m = \max_i \theta r_i, \quad (4)$$

which avoids numerical overflow. The first derivative is the tilted mean

$$\hat{\Lambda}'_{t,w}(\theta) = \sum_i p_i(\theta) r_i, \quad p_i(\theta) = \frac{\exp(\theta r_i)}{\sum_j \exp(\theta r_j)}. \quad (5)$$

The restricted empirical rate function is the Legendre-Fenchel transform [24]

$$\hat{I}_{t,w}(x) = \sup_{\theta \in \Theta} \{\theta x - \hat{\Lambda}_{t,w}(\theta)\}. \quad (6)$$

When the optimum is interior, θ^* satisfies

$$\hat{\Lambda}'_{t,w}(\theta^*) = x. \quad (7)$$

C. Interpretation under finite samples and dependence

The estimator is not intended to recover an asymptotic LDT rate for crypto returns, whose dependence, heavy tails, and regime shifts violate the clean theorem-level setting. We use $\hat{I}_{t,w}(x)$ as a finite-window convex stress coordinate: it measures the minimum exponential-tilting cost required to make a target average return x plausible under the recent empirical distribution. This is the same change-of-measure geometry that underlies exponential tilting in rare-event simulation [13] and the dual relation between entropic risk and relative-entropy penalties [14], [15]. A lower downside value of $\hat{I}$ therefore means that less distributional distortion is needed to move the window toward a drawdown-consistent state.

D. Effective Sample Size

Large tilts can be dominated by a few observations. We therefore compute

$$\text{ESS}_{t,w}(\theta) = \left(\sum_i p_i(\theta)^2 \right)^{-1} \quad (8)$$

and discard estimates with $\text{ESS}_{t,w}(\theta^*) < \tau_{\text{ESS}}$. Targets outside the empirical support of the rolling window are also marked as unavailable.

E. Drawdown-Consistent Thresholds

A cumulative drawdown of size δ over h days corresponds to

$$\frac{1}{h} \log \left(\frac{P_{t+h}}{P_t} \right) \leq \frac{\log(1 - \delta)}{h}. \quad (9)$$

Therefore the large-deviation threshold for a cumulative draw-down is the average log return

$$a_h(\delta) = \frac{\log(1 - \delta)}{h}. \quad (10)$$

This correction is important: using $I(-\delta)$ for an h -day cumulative drawdown mixes cumulative and per-period units.

F. Risk Features

The horizon-adjusted large-deviation proxy is

$$\text{LDP}_{t,w}(\delta, h) = \exp\{-h\hat{I}_{t,w}(a_h(\delta))\}. \quad (11)$$

It is an asymptotically motivated score, not a calibrated finite-sample probability.

The horizon-adjusted downside large-deviation risk score is

$$\text{HDLDR}_{t,w}(\delta, h) = \frac{1}{\epsilon + \hat{I}_{t,w}(a_h(\delta))}. \quad (12)$$

The reciprocal is used because the economic object of interest is fragility: risk rises when the rate-function barrier falls. The reciprocal makes larger values correspond to higher risk and improves interpretability in time-series plots. In the empirical implementation, $\epsilon = 10^{-6}$ is used only as a numerical regularizer to avoid division by zero when a rate estimate is extremely close to zero; ROC-AUC results are driven by rank variation in $\hat{I}_{t,w}(a_h(\delta))$, not by the regularization constant.

The interpretation of (11)–(12) is deliberately diagnostic. The estimator asks how expensive, in the recent empirical return distribution, it is to tilt the window toward an average return consistent with a cumulative drawdown. HDLDR is therefore a monotone transformation of a barrier estimate. It should be read as a state variable for downside fragility, while any use as a one-day VaR forecast requires an explicit calibration layer such as (15).

The downside-upside asymmetry gap is

$$\text{DAG}_{t,w}(\delta) = \hat{I}_{t,w}(\log(1 + \delta)) - \hat{I}_{t,w}(\log(1 - \delta)). \quad (13)$$

Unlike skewness, DAG compares the empirical large-deviation cost of equal upside and downside moves at the chosen threshold. Finally, rate-function collapse over lag k is

$$\text{RFC}_{t,w}(\delta) = \frac{\hat{I}_{t-k,w}(a_1(\delta)) - \hat{I}_{t,w}(a_1(\delta))}{\hat{I}_{t-k,w}(a_1(\delta))}. \quad (14)$$

Positive RFC means the downside barrier has fallen over the lookback period. The horizon-consistent RFC-accelerated variant used in the multivariate experiments replaces the fixed lag k with the 365-day rolling maximum of the same horizon-adjusted downside rate, so it measures collapse from the recent annual barrier peak rather than from a fixed calendar lag.

G. Calibrated HDLDR-to-VaR Map

To test whether the diagnostic score can also support a calibrated one-day tail forecast, we estimate a parsimonious HDLDR-to-VaR map. Let z_t be the expanding-window standardized value of $\log(1 + \text{HDLDR}_{t,w}(10\%, 7))$. The one-day 99% VaR forecast is

$$\text{VaR}_{t+1}^{\text{HDLDR}} = \exp\{\beta_{0,t} + \beta_{1,t}z_t\}. \quad (15)$$

Here $\text{VaR}_{t+1}^{\text{HDLDR}}$ denotes the predicted 99th-percentile loss (a positive number); the one-day 1% VaR violation occurs when $r_{t+1} < -\text{VaR}_{t+1}^{\text{HDLDR}}$. The parameters are re-estimated at each monthly walk-forward recalibration by minimizing lower-tail pinball loss at $\alpha = 0.01$ on past returns only. The exponential map is used to enforce a positive loss forecast; no claim is made that this log-linear form is the optimal VaR calibration. This makes the VaR experiment a calibrated forecasting test rather than a direct use of the large-deviation proxy as a probability.

Algorithm 1: Rolling Rate-Surface Estimation

Input: returns r_t , window w , thresholds δ , horizons h , tilt grid Θ , ESS cutoff τ_{ESS}

1. For each date t , set $\mathcal{W}_{t,w} = \{r_{t-w}, \dots, r_{t-1}\}$.
2. Estimate $\hat{\Lambda}_{t,w}(\theta)$ with log-sum-exp.
3. For each target $a_h(\delta)$, solve (6).
4. Compute ESS(θ^*) and discard unstable estimates.
5. Store $\hat{I}$, HDLDR, LDP, DAG, and RFC.
6. Optionally calibrate (15) by past pinball loss.
7. Evaluate scores only in walk-forward out-of-sample tests.

Main setting: $w = 180$, $\delta \in \{3\%, 5\%, 10\%\}$, $h \in \{1, 7, 14, 30\}$, $\Theta = [-250, 250]$, $\tau_{\text{ESS}} = 10$.

IV. EXPERIMENTAL SETUP

A. Data

We use BTCUSDT and ETHUSDT daily Binance spot and USD-M futures data from 2019-01-01 to 2026-05-09 [25], collected from `/api/v3/klines`, `/fapi/v1/klines`, and `/fapi/v1/fundingRate`; open interest and long-short positioning come from Binance Data Vision [26]. The empirical feature freeze is 2026-05-09. The daily sample contains 2685 log returns per asset; funding rates cover 2434 BTC and 2356 ETH days; open interest covers 1953 BTC days (from 2021-01-01) and 1621 ETH days (from 2021-12-01). Table I summarizes the source windows and validation roles. For the short-sample real-liquidation extension we use Crypto-HFTData Binance Futures liquidation records [27] aggregated to one-hour buy- and sell-side notional; licensed liquidation outcomes are observed through 2026-05-12 23:00 UTC to construct forward 72-hour labels whose feature timestamps end on 2026-05-09. Full historical USD-M liquidation snapshots are not available in the public Binance archive, so the intraday analysis is reported separately and is not used as full-cycle liquidation-cascade evidence. All experiments use Python 3.11 with numpy 1.26, scipy 1.11, scikit-learn 1.3, xgboost 2.0, and arch 6.2; the random seed is fixed at 42. Code, features, and intermediate outputs will be released through an anonymized reproducibility archive for peer review.

B. Validation Design

The main rate surfaces use $w = 180$, $\delta \in \{3\%, 5\%, 10\%\}$, $h \in \{1, 7, 14, 30\}$, $\Theta = [-250, 250]$, $\tau_{\text{ESS}} = 10$. VaR models are recalibrated every 30 days in an expanding-window walk-forward design beginning in 2021. Drawdown prediction

TABLE I
EMPIRICAL DATA WINDOWS AND VALIDATION ROLES

Layer	Source window	Validation role
Daily returns/funding	2019-01-01–2026-05-09	Main VaR and drawdown tests; daily out-of-sample evaluation starts on 2021-01-01.
Daily futures OL/positioning	BTC: 2021-01-01–2026-05-09; ETH: 2021-12-01–2026-05-09	Derivatives covariates and funding/open-interest regime summaries.
One-hour OHLCV and OL frame	2021-01-01 00:00–2026-05-09 00:00	Intraday open-interest-shock control; out-of-sample evaluation starts on 2021-07-01.
One-hour licensed liquidations	BTC: 2025-06-28 05:00–2026-05-12 23:00; ETH: 2025-06-29 00:00–2026-05-12 23:00	Short-sample real-liquidation-pressure test; contiguous hourly source rows are 7651/7651 for BTC and 7632/7632 for ETH. Clean modeling runs from 2025-06-29 00:00 to 2026-05-09 00:00, and 72h prediction rows run from 2025-09-27 00:00 to 2026-05-09 00:00 after 90 days of training history.

is evaluated out of sample from 2021-01-01 onward. Benchmarks include historical, Gaussian, Student-t, EWMA, GARCH(1,1), GARCH-t, EVT/POT, and HDLDR-calibrated VaR models. Robustness checks cover $w \in \{30, 90, 180, 365\}$, $|\Theta| \in \{150, 250, 400\}$, and $\tau_{\text{ESS}} \in \{5, 10, 20\}$. The out-of-sample period starts in 2021 so the March 2020 COVID crash informs the training history rather than serving as a test event. Pre-specification. The following targets were fixed before out-of-sample evaluation: (i) BTC 30-day drawdown with $\delta = 20\%$ as the primary daily drawdown target, motivated by the medium-horizon framing in Section III; and (ii) the liquidation-pressure targets at 6, 24, and 72 hours, with total liquidation targets at all three horizons and sell-side variants at 24 and 72 hours. Other asset, horizon, and threshold combinations are reported as exploratory. We do not apply formal family-wise correction; instead, dependence-robust block bootstrap is treated as the binding inference for primary claims, and exploratory results are interpreted as hypothesis-generating. The intraday test uses one-hour USD-M futures returns, open-interest metrics, and liquidation flows. Rate functions are estimated on 168-hour and 720-hour windows with 24-hour and 72-hour drawdown-equivalent targets; continuous targets are pre-specified as $\log(1 + \text{future liquidation USD})$ over 6, 24, and 72 hours, with sell-side variants at 24 and 72 hours. The conventional benchmark includes realized and downside volatility, skewness, kurtosis, momentum, volume, taker imbalance, open-interest changes, trailing liquidation pressure, and liquidation imbalance/z-scores. Walk-forward XGBoost and ridge models are updated every 24 hours with at least 90 days of training history, yielding 5377 out-of-sample hourly predictions per asset-target (2025-09-27 to 2026-05-09). Incremental gains are evaluated with paired block bootstraps at 24, 72, 168, and 336-hour blocks; Table VII reports the conservative 336-hour inference. All XGBoost models use fixed hyperparameters with no test-set tuning (daily: $n_{\text{estimators}}=100$, $\text{max_depth}=3$, $\text{learning_rate}=0.05$; intraday: $n_{\text{estimators}}=60$,

TABLE II
DESCRIPTIVE STATISTICS OF DAILY CRYPTOCURRENCY RETURNS

	BTC	ETH
Period	2019-01-01–2026-05-09	2019-01-01–2026-05-09
Observations	2685	2685
Mean return (% p.a.)	41.5445	38.2530
Volatility (% p.a.)	63.1744	83.0450
Skewness	-1.2816	-1.0771
Excess kurtosis	22.9171	16.4789
Min daily log-return (%)	-50.2607	-59.0534
Max daily log-return (%)	17.8449	23.3750
Historical VaR 99% (%)	9.2370	12.3670
Historical CVaR 99% (%)	13.3607	17.8696
Jarque-Bera p-value	0.0000	0.0000
ARCH(5) p-value	0.0000	0.0000

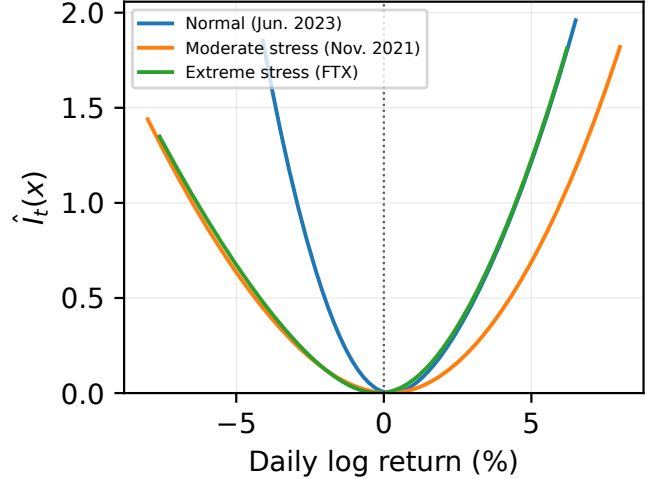


Fig. 1. Empirical BTC rate functions in three regimes. A flatter downside curve indicates a lower empirical barrier against negative fluctuations.

$\text{learning_rate}=0.04$; $\text{shared_subsample}=0.8$, $\text{colsample_bytree}=0.8$, $\text{min_child_weight}=5$, $\text{random_state}=42$). The training window expands by 30 days at each step with a minimum of 730 observations. Feature importance is measured by mean absolute SHAP values averaged across walk-forward folds.

V. RESULTS

A. Descriptive Statistics

Table II reports strong non-Gaussianity. Both assets have negative skewness, very high excess kurtosis, and significant ARCH effects. The one-day COVID crash contributes the most extreme daily observations, but the Jarque-Bera and ARCH tests confirm that non-normality and volatility clustering are not isolated artifacts.

B. Rate-Surface Geometry

Fig. 1 shows three BTC rate functions. The normal 2023 window has a steeper downside curve and a narrower empirical support, while stress windows display flatter downside geometry. This is the key visual object of the paper: the surface represents a local stability landscape, and a lower downside branch means a drawdown state is closer in exponential-tilting distance.

TABLE III
THRESHOLD CORRECTION: CORRECTED $a_h(\delta)$ VS. NAÏVE $x = -\delta$

Asset	Target	AUC corr.	AUC naïve	Valid obs.
BTC	7d DD 10%	0.5703	—	1955 / 29
BTC	30d DD 20%	0.6312	—	1955 / 29
ETH	7d DD 10%	0.5503	0.4794	1955 / 475
ETH	30d DD 20%	0.5556	0.4489	1955 / 475

BTC naïve: non-computable (ESS; $x = -\delta$ outside support).
 $\delta = 10\%$, $w = 180$; mismatch factor for 30d DD 20%: $26.9\times$.

TABLE IV
SELECTED OUT-OF-SAMPLE DRAWDOWN PREDICTION

Asset	Target	Event rate	RV AUC	HDLDR AUC	Best LDT AUC
BTC	7d DD 10%	0.106	0.604	0.570	0.580
BTC	30d DD 20%	0.109	0.530	0.631	0.659
ETH	7d DD 10%	0.181	0.594	0.550	0.534
ETH	30d DD 20%	0.213	0.586	0.556	0.600

RV = realized volatility; Best LDT is walk-forward logistic or XGBoost LDT model.

Fig. 2 plots BTC price and HDLDR(10%, 7d). The score rises sharply around the COVID crash and remains elevated in the 2022 stress regime around Terra/Luna and FTX. This supports the interpretation of HDLDR as a stress-state diagnostic, while also showing that the measure is not a clean point predictor of every crash.

C. Threshold Correction Validation

Table III compares HDLDR using the horizon-consistent threshold $a_h(\delta)$ defined in (10) against a naïve version that uses $x = -\delta$ directly. For a 30-day drawdown of 20%, the corrected threshold is $a_{30}(0.20) = \log(0.80)/30 \approx -0.00744$ per day, whereas the naïve threshold is -0.20 , a factor of $26.9\times$ larger in magnitude. For BTC, the naïve version produces only 29 valid out-of-sample estimates versus 1955 for the corrected version, because $x = -0.20$ lies outside the rolling daily-return support on most windows and triggers the ESS filter. For ETH, the naïve version yields 475 valid observations and AUC improvements of $+0.07$ – $+0.11$ from the correction. The unit alignment in (10) is therefore a methodological requirement, not a cosmetic rescaling.

D. Drawdown Prediction as Cross-Regime Diagnostic

Table IV reports ROC-AUC for future drawdown events. For 7-day drawdowns of at least 10%, realized volatility is stronger than HDLDR for both BTC and ETH. For 30-day BTC drawdowns of at least 20%, HDLDR reaches 0.631 AUC versus 0.529 for realized volatility, and a walk-forward XGBoost model using HDLDR, DAG_{3%}, horizon-consistent RFC, and regime features reaches 0.659. This is the daily result that supports the diagnostic framing: rate surfaces separate persistent BTC stress regimes better than they time near-term crash events. ETH and short-horizon labels remain weak settings for the method. DAG at $\delta = 5\%$ does not provide incremental ranking signal in this design; only DAG_{3%} is used in downstream LDT models.

Table V reports DeLong tests for key AUC comparisons. The BTC 30-day HDLDR result is significant under the parametric correlated-AUC test ($z = 4.71$, $p = 2.5 \times 10^{-6}$), but the dependence-robust block bootstrap is the binding inference: at

TABLE V
STATISTICAL SIGNIFICANCE OF AUC DIFFERENCES (DELONG TEST)

Asset	Target	Model	AUC (Model)	AUC (Realized Vol)	Δ AUC	z	p
BTC	30d DD 20%	HDLDR	0.6312	0.5295	+0.1017	4.71	2.5×10^{-6}
BTC	7d DD 10%	HDLDR	0.5703	0.6039	−0.0336	−1.54	0.123
ETH	30d DD 20%	HDLDR	0.5556	0.5858	−0.0302	−1.87	0.062
ETH	7d DD 10%	HDLDR	0.5503	0.5940	−0.0437	−2.55	0.011

DeLong (1988) correlated-AUC test; baseline is Realized Vol 30d in all rows.

TABLE VI
XGBOOST ABLATION: 30-DAY $\geq 20\%$ DRAWDOWN

Feature Set	BTC 30d DD $\geq 20\%$		ETH 30d DD $\geq 20\%$	
	ROC-AUC	PR-AUC	ROC-AUC	PR-AUC
Realized vol only	0.5702	0.1269	0.4637	0.1908
HDLDR only	0.6368	0.2144	0.6127	0.2557
DAG + RFC only	0.6667	0.1777	0.5986	0.3011
Realized vol + HDLDR [†]	0.6152	0.1893	0.5599	0.2266
Realized vol + LDT feats	0.6826	0.2820	0.5720	0.2758
Full model (all features)	0.6940	0.2764	0.5663	0.2783

Walk-forward XGBoost (rolling 730-day window, test 2021+). LDT feats: HDLDR, DAG_{3%}, RFC-acc. Full model adds OI change and funding regime flag. Mean SHAP (BTC): RFC-acc (0.934) > DAG_{3%} (0.548) > HDLDR (0.434) > vol (0.347).
[†]HDLDR only > Realized vol + HDLDR: adding a weak predictor (vol AUC=0.570 standalone) consumes tree-split budget without adding signal, reducing net performance. The full model recovers because DAG and RFC are themselves informative.

a 21-day block, the 95% CI includes zero ($[-0.006, 0.259]$) and $p = 0.064$. Longer blocks of 30, 45, and 60 days give $p = 0.096$, 0.112, and 0.152. The result is economically meaningful and directionally stable, but marginal once serial clustering of drawdown labels is respected.

Table VI reports a walk-forward XGBoost ablation. Adding HDLDR to realized volatility raises BTC AUC from 0.570 to 0.615 and ETH from 0.464 to 0.560; the full LDT model reaches 0.694 for BTC. By mean SHAP, RFC-accelerated is the largest contributor, followed by DAG_{3%} and HDLDR. Against a stronger kitchen-sink XGBoost benchmark using realized volatility, downside volatility, skewness, kurtosis, rolling VaR/CVaR, trailing drawdown, momentum, funding, basis, and open-interest features, the incremental daily LDT gain is modest and not robust: BTC AUC moves from 0.727 to 0.732 (30-day block-bootstrap $p = 0.904$), and ETH from 0.579 to 0.607 ($p = 0.560$). SHAP values should therefore be read as model inspection rather than causal feature importance.

E. Temporal Stability and Cross-Regime Separability

Per-year HDLDR AUCs for the BTC 30-day target fall below 0.5 in every individual subperiod: the Full OOS AUC of 0.631 arises because HDLDR scores are systematically elevated in 2021–2022 (158 BTC events) relative to 2024–2026 (51 events; no breaches in 2023), creating cross-regime separability that within-year evaluation cannot capture. This is a Simpson-type ranking effect: pooled ROC-AUC is a probability over all positive-negative pairs, so between-regime pairs dominate the pooled ranking even when within-year rankings are weak. The rate surface is therefore best interpreted as a slowly varying diagnostic of persistent cross-regime fragility, not a within-regime event timer. Excluding 2022 entirely, BTC HDLDR remains above realized volatility (0.567 vs. 0.547), so the result is not driven by a single year; for ETH, excluding 2022 yields HDLDR 0.514 vs. realized volatility 0.634, so the ETH advantage is absent outside the 2022 stress cluster.

F. One-Hour Liquidation Pressure

Table VII reports the strongest empirical evidence in the paper. The target is future realized liquidation pressure,

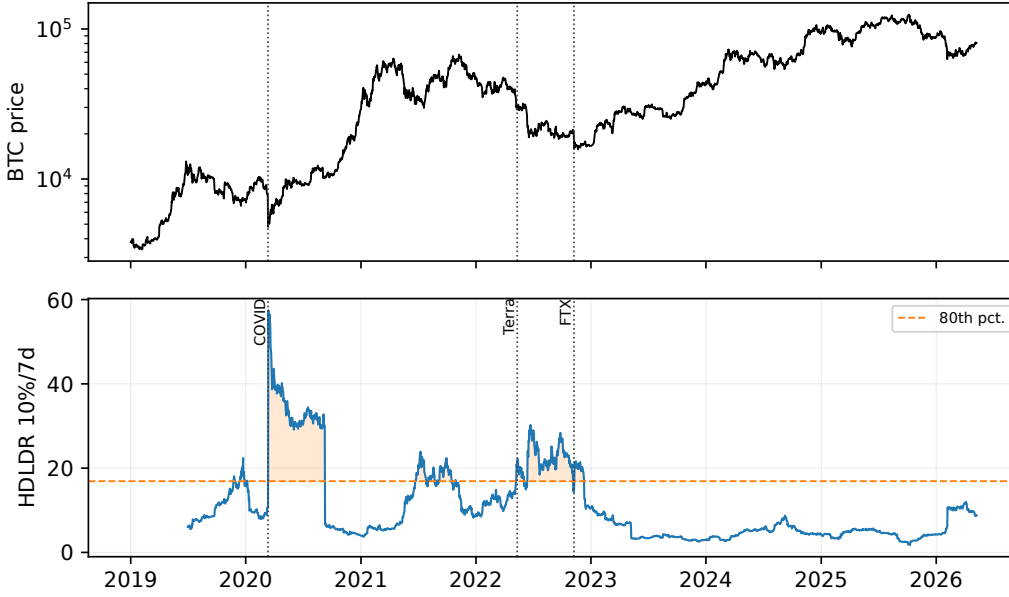


Fig. 2. BTC price and horizon-adjusted downside large-deviation risk. The lower panel reports HDLDR(10%, 7d) with major stress events marked; the 80th-percentile line is a visual reference, not a selected decision threshold.

TABLE VII
ONE-HOUR LIQUIDATION-PRESSURE TEST: XGBOOST Δ IC BY HORIZON

Asset	Target	Δ IC 6h	Δ IC 24h	Δ IC 72h	IC +LDT 72h	336h p
BTC	Total liq.	+0.011	+0.057	+0.211	0.629	< 0.005
BTC	Sell liq.	—	+0.029	+0.277	0.592	< 0.005
ETH	Total liq.	+0.004	+0.063	+0.186	0.622	< 0.005
ETH	Sell liq.	—	+0.035	+0.203	0.567	0.015

Δ IC is the gain from adding 1h LDT features to conventional intraday, OI, and trailing-liquidation features. Sell-side targets were pre-specified at 24h and 72h. OOS $N = 5377$ hourly predictions per row.

$\log(1 + \text{liquidation USD})$, rather than a binary daily draw-down label. This target is closer to the economic mechanism: downside fragility should matter most when it interacts with leverage, open interest, and forced liquidation flows. The 72-hour gains are large for both assets and both total and sell-side pressure, while 6-hour increments are near zero and 24-hour increments are positive but smaller. This horizon pattern supports a persistent-fragility interpretation rather than point-event timing.

For BTC total liquidation pressure, adding one-hour LDT features to the conventional benchmark raises Spearman IC from 0.418 to 0.629; for sell-side pressure, from 0.316 to 0.592. ETH shows the same direction: 0.436 to 0.622 for total pressure and 0.363 to 0.567 for sell-side pressure. The paired 336-hour block-bootstrap intervals remain positive: [0.106, 0.369], [0.181, 0.445], [0.086, 0.274], and [0.063, 0.311], respectively. The result is short-sample evidence from one venue, but it is materially stronger than the daily drawdown classification evidence and is aligned with the liquidation mechanism that motivates the diagnostic.

To check whether this is an artifact of the shorter liquidation sample, we run the same one-hour feature pipeline on the longer 2021-07-01–2026-05-09 open-interest shock targets. There, incremental LDT gains are small and not robust under 336-hour block bootstraps (e.g., BTC 72-hour absolute OI

TABLE VIII
SELECTED OUT-OF-SAMPLE VaR BACKTESTS, 99% VaR

Asset	Model	Obs.	Viol.	Viol. Rate	Kupiec p	Christ. p	QLoss
BTC	Gaussian VaR	1955	20	0.0102	0.9188	0.1994	0.0011
BTC	GARCH-t	1955	2	0.0010	0.0000	0.9490	0.0016
BTC	EVT/POT VaR	1955	9	0.0046	0.0073	0.7729	0.0012
BTC	HDLDR-Calibrated VaR	1955	14	0.0072	0.1839	0.6531	0.0012
ETH	Gaussian VaR	1955	23	0.0118	0.4455	0.2729	0.0016
ETH	GARCH-t	1955	6	0.0031	0.0003	0.0116	0.0019
ETH	EVT/POT VaR	1955	13	0.0066	0.1129	0.0736	0.0016
ETH	HDLDR-Calibrated VaR	1955	14	0.0072	0.1839	0.0875	0.0016

shocks: IC moves from 0.280 to 0.304, $p = 0.315$). The favorable intraday evidence is therefore specific to real liquidation-pressure targets, not a generic result of moving to one-hour frequency. Within the LDT block, leave-one-out diagnostics identify $\text{RFC}_{3\%, 24h, 720h}$ as the strongest individual contributor across all four asset-target rows, although single-feature gains do not reproduce the full block gains. A formal placebo test replacing LDT features with autocorrelation-matched noise is left for future work; the leave-one-LDT-feature-out diagnostic and the open-interest-shock counterfactual provide partial controls against specification mining.

G. VaR Calibration

HDLDR can be calibrated into a one-day 99% VaR forecast using (15). As Table VIII shows, the HDLDR-calibrated VaR passes unconditional coverage for BTC and ETH, but it does not improve quantile loss relative to Gaussian VaR; for ETH, conditional coverage is also borderline ($p = 0.0875$). This is a consistency check on calibration, not a contribution claim.

H. Robustness

We re-estimate HDLDR for $w \in \{30, 90, 180, 365\}$ on the main daily targets. For BTC 30-day drawdowns of at least 20%, HDLDR AUC remains above 0.61 for all $w \geq 90$ (0.644,

0.631, 0.612 for $w = 90, 180, 365$). ETH shows a faster regime profile: AUC=0.649 at $w = 30$ versus 0.556 at $w = 180$, suggesting that ETH benefits from shorter estimation windows. Across $|\Theta| \in \{150, 250, 400\}$ and $\tau_{\text{ESS}} \in \{5, 10, 20\}$, AUC for the BTC and ETH 30-day targets is unchanged (0.631, 0.556), because the optimized thresholds lie inside the rolling return support and the ESS constraint does not bind at $w = 180$.

VI. DISCUSSION

The rate surface is not a replacement for volatility, GARCH, EVT, or calibrated VaR. Its strongest daily univariate result appears for severe 30-day BTC drawdowns rather than 7-day drawdowns, matching the construction of the statistic: a multi-month rolling window estimates local distributional geometry suited to detecting persistent fragility, not to timing abrupt one-week shocks. The parametric DeLong test is significant for the primary BTC 30-day result ($p < 0.001$), but the dependence-robust block bootstrap ($p = 0.064$) is the binding inference once serial clustering of drawdown labels is taken into account. Within the kitchen-sink XGBoost benchmark, the incremental daily LDT contribution is feature- and asset-specific and not robust under block bootstrap. The one-hour liquidation extension is consistent with a microstructure interpretation: rate-surface dynamics add information when the target is realized liquidation pressure and the feature frequency is aligned with leverage adjustment, with the 72-hour signal stronger than the 6-hour signal. Negative funding in this sample is more often associated with low downside barriers than positive funding, indicating that funding sign alone is an incomplete proxy for leverage fragility.

Limitations. Coverage is limited to BTC and ETH; the BTC result does not establish generality. Future drawdown labels are serially correlated, so the block bootstrap is treated as the binding inference even when DeLong is significant. The main sample is daily and the one-hour liquidation extension covers only mid-2025–2026 because full historical USD-M liquidation snapshots are not available in the public Binance archive. Direct comparison against Markov-switching regime classifiers is left for future work; the current benchmark layer focuses on volatility, EVT, and a multivariate XGBoost kitchen-sink. Future work should also add a formal placebo that substitutes LDT features with autocorrelation-matched noise, bootstrap uncertainty bands for the rate surface itself, and benchmark production latency.

VII. CONCLUSION

Rolling Legendre-Fenchel rate surfaces are best viewed as diagnostic coordinates for crypto fragility, not as universal crash predictors. The horizon-consistent threshold correction is a replicable methodological contribution: without it, cumulative drawdown targets are placed on the wrong per-period scale and the BTC estimator becomes non-computable at the main parameters. The one-hour liquidation-pressure extension provides the clearest mechanistic evidence, with rate-surface features improving Spearman IC over a strong

intraday benchmark and retaining positive 336-hour bootstrap intervals for BTC and ETH at the 72-hour horizon. The daily drawdown evidence supports cross-regime separability for severe medium-horizon BTC stress, but it is marginal under dependence-robust bootstrap inference and weaker for ETH and short horizons. The appropriate use is therefore complementary: rate surfaces add an interpretable fragility coordinate alongside volatility, EVT, GARCH, and calibrated VaR models.

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